

Operational Roadmap for the Mechanical Power Module (MPM)

Takeaway: This roadmap lays out the full operational trajectory of the MPM—from prototype to national deployment—structured as a standards-grade, multi-phase plan. It integrates engineering, logistics, regulatory alignment, consortium governance, and industrial scaling. Guided Links are embedded so you can expand any component.

1. Strategic Intent

The MPM is a universal mechanical-power primitive. Its operational roadmap must ensure:

- Repeatable, safe, interoperable mechanical-power delivery
- Multi-vendor manufacturing and host-system compatibility
- Logistics workflows that treat the MPM as a containerized asset
- Regulatory alignment across DOE, FERC, ASME, and state agencies
- Consortium governance that preserves competitive neutrality

This roadmap is the operational spine for MPM Standard v1.0 and the MPM Consortium launch.

2. Roadmap Phases

The roadmap is structured into **seven operational phases**, each with clear objectives, deliverables, and transition criteria.

Phase 1 — Foundational Engineering & Standardization

Focus: Establish the technical foundation.

- Finalize **geometry envelope** and mechanical interface tolerances
- Define **safety envelope** and interlock requirements
- Publish **MPM Standard v0.9** for pilot fabrication
- Build reference implementations (module, host adapter, latch)
- Establish NIST-traceable measurement requirements

Deliverables:

- MPM Standard v0.9
- Reference module (R1)
- Host-system adapter set
- Safety-interlock reference design

Phase 2 — Pilot Fabrication & Host-System Integration

Focus: Produce pilot-grade modules and retrofit host systems.

- Fabricate pilot modules to v0.9 tolerances
- Integrate host-system adapters into industrial equipment
- Validate mechanical coupling, alignment, retention
- Instrument telemetry and compliance logging

Deliverables:

- Pilot module batch
- Host-system integration packet
- Telemetry schema v1.0

Phase 3 — Logistics Infrastructure Deployment

Focus: Treat the MPM as a logistics primitive.

- Deploy swap stations at pilot sites
- Install transport frames and handling systems
- Train operators in swap-out and safety procedures
- Validate logistics repeatability

Deliverables:

- Swap-station readiness report
- Operator certification
- Logistics workflow documentation

Phase 4 — Controlled Pilot Deployment

Focus: Introduce the MPM into real-world duty cycles.

- Run low-load and full-load mechanical tests
- Validate torque, speed, thermal, vibration envelopes
- Conduct swap-out cycles (target: <15 minutes)
- Test failure modes and emergency stop logic

Deliverables:

- Pilot performance dataset
- Failure-mode validation report
- Swap-out time analysis

Phase 5 — Data Analysis & Standards Refinement

Focus: Convert pilot data into standards-grade improvements.

- Analyze interface wear, alignment drift, latch reliability
- Update geometry tolerances and safety requirements
- Refine telemetry schema and compliance thresholds
- Publish MPM Standard v1.0

Deliverables:

- Standards revision packet
- MPM Standard v1.0
- Updated compliance test suite

Phase 6 — Consortium Launch & Multi-Vendor Scaling

Focus: Establish governance and expand manufacturing.

- Launch the **MPM Consortium**
- Formalize compliance certification workflow
- Onboard manufacturers, OEMs, logistics providers
- Publish competitive-neutrality charter
- Begin multi-vendor module production

Deliverables:

- Consortium charter
- Compliance certification program
- Multi-vendor interoperability roadmap

Phase 7 — National & International Deployment

Focus: Scale the MPM across industries and borders.

- Align with **DOE, FERC, ASME, NIST**, and state agencies
- Deploy MPM modules across water, industrial, and mobile systems
- Expand swap-station networks
- Begin ISO/IEC harmonization for global adoption

Deliverables:

- Regulatory alignment packet
- National deployment plan
- International standards harmonization roadmap

3. Operational Architecture

The roadmap is supported by a stable operational architecture.

A. Engineering Backbone

- Open mechanical interface
- Standard geometry envelope
- Redundant safety interlocks
- NIST-traceable measurement framework

B. Logistics Backbone

- Swap stations
- Transport frames
- Handling protocols
- Operator certification

C. Governance Backbone

- Standards Board
- Compliance Authority
- Deployment Council
- Competitive-neutrality safeguards

D. Regulatory Backbone

- DOE CRADA alignment
- FERC permitting alignment
- ASME mechanical standards
- State safety codes

4. Key Milestones

- **MPM Standard v0.9** published
- Pilot modules fabricated
- Swap stations deployed
- Pilot data collected
- **MPM Standard v1.0** finalized
- Consortium launched
- Multi-vendor manufacturing begins
- National deployment initiated

5. Success Criteria

The MPM roadmap is successful when:

- Modules operate across multiple host systems without modification
- Swap-out time is consistently <15 minutes
- Safety envelope shows zero uncontrolled disconnects
- Telemetry supports real-time compliance
- Multiple vendors manufacture compliant modules
- Regulators recognize the MPM as a non-proprietary standard